STACKS, DEFORMATION THEORY: LEMMA-THICKENING-OVER-THICKENING-SPACE-QUASI-COHERENT

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 5094-5101 of sources/stacks-defos/pinned-original-source.tex; proof span is lines 5103-5120, the complete author proof (16 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 37 of queue_defos.json (pb=822, stmt 5094-5101, proof 5103-5120).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 08UF: sources/stacks-defos/pinned-original-source.tex, source lines 4101–4110. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0D3P: sources/stacks-defos/pinned-original-source.tex, source lines 5045–5051. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CR3: sources/stacks-defos/pinned-original-source.tex, source lines 5120–5120. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *In the situation above assume that X is quasi-compact and quasi-separated and that $DQ_X(\mathcal{F}) \to DQ_X(\mathcal{G})$ (Derived Categories of Spaces, Section 0CR3) is an isomorphism. Then the functor FT is an equivalence of categories.*

Proof. A solution of (08UF) for $\mathcal{F}$ in particular gives an extension of $f^{-1}\mathcal{O}_{B'}$-algebras

$$0 \to \mathcal{F} \to \mathcal{O}' \to \mathcal{O}_X \to 0$$

where $\mathcal{F}$ is an ideal of square zero. Similarly for $\mathcal{G}$. Moreover, given such an extension, we obtain a map $c_{\mathcal{O}'} : f^{-1}\mathcal{J} \to \mathcal{F}$. Thus we are looking at the full subcategory of such extensions of $f^{-1}\mathcal{O}_{B'}$-algebras with $c = c_{\mathcal{O}'}$. Clearly, if $\mathcal{O}'' =$

Date: stacks-project:defos.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

$F(\mathcal{O}')$ where F is the equivalence of Lemma 0D3P (applied to $X \to B'$ this time), then $c_{\mathcal{O}''}$ is the composition of $c_{\mathcal{O}'}$ and the map $\mathcal{F} \to \mathcal{G}$. This proves the lemma. $\square$